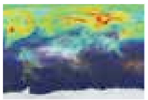
FES e-ink watch
FES Watch lets its wearers change the design of the watch and also its entire strap with a simple button press **http://dgit.in/FESeInk**



Carbon Dioxide emissions
Check out this amazing visualization of the Carbon Dioxide emissions and its effect on the planet **http://dgit.in/CO2Emission**

almost junk-like objects that he found lying around. We always took extra comfort and felt a sense of security in the fact that to come up with innovative solutions, one needs to possess the sort of creativity that couldn't be coded on a computer. Even the thought of a robot solving a problem with a MacGyver-like approach was laughable.

Hopefully, it shouldn't alarm you that robots are now starting to do exactly this! In a video submission to the IEEE International Conference on Robotics and Automation 2004, Georgia Tech's 'Golem Krang Robot' was shown to use the very objects that were posed to it as obstacles to perform the specific task required of it, i.e. rescue a trapped person stuck behind the door blocked by the above mentioned obstacles. The robot used objects in the environment as tools to clear other objects that blocked access to the door, which it then opened to rescue the 'victim'.

Just a few years ago, this sort of situation would have only been believable in a Sci-Fi movie. However, in this scenario, the robot is programmed and designed on the basis of a novel concept called 'Environment Aware Planning' (or 'ETAP'), which lets it analyse available objects and resources in the environment and subsequently implement a plan of action that optimally uses these objects to complete the task. This type of technology can be used to design more self-aware robots that can be deployed in complex search-and-rescue missions. It can also be used to solve constructional or logistics-related problems, which would take longer if they were carried out by humans.

The real question is: Can it win at Jenga?

Wearable robotech

For a country like Japan, which prides itself on the score of technological advancements it has introduced world over, it comes as no surprise that the government will be increasing investments in healthcare robots to assist the ageing population (over one-quarter of Japan's population is aged 65 years and older). It will do this by extending financial subsidies to help firms develop inexpensive nursing care robots. The government aims to produce healthcare robots that will be ready for the market by 2016 at a selling cost of nearly $1,000 (₹61,835) a unit.

Fortunately, there's been considerable progress in the field of bionics and biomedical engineering. We now have extremely advanced and highly functional 'wearable robots' or 'exoskeletons', which are essentially mechanoelectrical suits worn by disabled individuals to assist them with basic biomechanical functions. Take, for example, the Robot Suit HAL – a futuristic suit used in the medical rehabilitation of patients who've suffered a stroke or spinal cord injury. The suit captures human nerve signals sent from the brain to the muscles and uses these signals to control the movement of the joints in the suit.

Alternatively, a DARPA-funded project exists in which engineers are designing a robotic exosuit to improve the endurance of soldiers, who have to routinely carry heavy backpacks over long distances. Known as the 'Mobility Enhancing Soft Exosuit', it's made using lightweight and flexible materials and aims to prevent and reduce musculoskeletal injuries suffered by military personnel. Similar to the HAL suit, a working prototype comprises a series of webbing straps that are positioned around the lower half of the body. This prototype contains a low-power microprocessor and network of supple strain sensors that act as the respective 'brain' and 'nervous system' of the Soft Exosuit.

Pictured here: Tony Starks's biggest fan

Automatonic swarm-bots

The whole concept of 'swarm robotics' is relatively new in terms of general knowledge of the masses. Video gamers, however, will identify with the concept of 'swarming', which is referenced in the popular *StarCraft* series of real-time strategy games. Much like in the game, the term 'swarm' in robotics refers to a collection of robots that move and behave in a predictive and collective pattern in order to accomplish tasks that couldn't have been achieved by a single robot.

It's one thing to imagine a single robot behaving autonomously. A huge swarm of similar robots behaving with a potentially dangerous group-hive mentality is a different beast altogether. The US Navy recently dabbled with the idea of using 'swarming' drone boats programmed to act autonomously and with minimal human guidance. In a recent example of a naval training exercise with these drones, a large ship (referred to in the exercise as the 'high-value unit' or 'HVU') was making its way